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MID TERM EXAMINATION

B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
Second Semester, March, 2026

Paper Code: BS 106**Subject: Applied Physics-II****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Max. Marks	CO(s)
Question - 1			
1 (a)	Using exchange symmetry of wave function, show that Bosons do not obey Pauli Exclusion Principle.	2.5	CO2
1 (b)	An eigen function of an operator $\frac{d^2}{dx^2}$ is $\psi = e^{ax}$. Find the corresponding eigen value.	2.5	CO1
1 (c)	An electron has a speed of $2 \times 10^4 \text{ m s}^{-1}$ within the accuracy 0.01%. Calculate the uncertainty in the position of the electron.	2.5	CO1
1 (d)	Calculate the root mean square speed of oxygen molecules at 0°C. Given mass of $O_2 = 5.31 \times 10^{-26} \text{ kg}$.	2.5	CO2
Question - 2			
2 (a)	The wave function of a certain particle is $\psi = A \cos^2 x$ for $-\pi/2 < x < \pi/2$. Find the value of A and the probability that the particle be found between $x = 0$ and $x = \pi/4$.	5	CO1
2 (b)	An electron is in a box of width 2 Å , calculate the separation between the lowest two levels in eV.	2	CO1
2 (c)	Write the necessary conditions for quantum tunnelling. Calculate the transmission probability for an electron of energy 2 eV when incident upon a rectangular potential barrier of height 6 eV and width 0.50 nm.	3	CO1
Question - 3			
3 (a)	Use the distribution function to show that the Bose-Einstein (BE) and Fermi-Dirac (FD) distributions at very high temperature reduce to the Maxwell-Boltzmann (MB) distribution.	4	CO2
3 (b)	Explain Fermi-Dirac distribution function. Plot this function as a function of energy for various temperatures including 0 K.	3	CO2
3 (c)	(i) Show that at $E = E_F$ and $T \neq 0 \text{ K}$, the occupancy is equal to 0.5. (ii) Show that at $E < E_F$ and $T = 0 \text{ K}$, the occupancy of state is 1, and at $E > E_F$ and $T = 0 \text{ K}$, the occupancy of state is zero.	3	CO2
Question - 4			
4 (a)	Evaluate the expectation value of the position and the momentum of a particle trapped in a box of width a.	5	CO1
4 (b)	Distribute 5 particles in five energy states in accordance with MB, BE and FD statistics.	3	CO2
4 (c)	Calculate the wavelength of solar radiation to which the human eye is most sensitive. Assume the temperature of the sun to be 5700 K.	2	CO2

****** End of Question Paper ******